

Nova: A new Internet protocol stack

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ABSTRACT

Imagine a new Internet that requires only minimal configuration and management of routers and clients and where the user pays the connectivity doing real-time nano-payments to the network and where each node participating in that routing process receive his share. This paper presents an informal exposition of the design rationale behind the development of Nova, a replacement of Internet protocols stack, that make this value proposition a reality, solving the recognized problems of TCP-IP and creating economic incentives for a more distributed network topology and the introduction of new stack enabled network hardware. Nova solves at its root the net neutrality problem in a market-friendly way and guarantees an unprecedented level of confidentiality, implementing the concept of "privacy by design". It uses technologies that until now had been applied only to P2P and blockchain worlds to disintermediate the Internet turning it into a distributed trust-less network, achieving a reduction in costs of deploying and maintenance of the routing infrastructure, disrupting for the better the business model for all the actors of the Internet. Nova provides a gradual and progressive way to its adoption without points of systemic singularity, creating the new Internet in the shell of the old.

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Please note: the key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC2119[1].

1 INTRODUCTION

The persistent success of the TCP-IP arises from the so-called "end-to-end principle". As noticed by Courcoubetis and Weber[2]:

"The end-to-end principle says that if such a function is rarely invoked, and is not common to all data traffic, then it is better to install it at the edge of the network, rather than in each link of the network. The reduction in redundancy results in a simpler network core. Customer devices at the edges of the network must provide whatever extra functionality is needed to support the quality requirements of a given application. The fact that the Internet is stupid is one of the major reasons for its success."

Later Antonopoulos[3] highlights:

*"[TCP-IP] doesn't offer services. It doesn't make decisions about content. It doesn't distinguish between photos and text, video and audio. It doesn't have a list of approved applications. It doesn't even distinguish between client and server, user and host, or an individual versus a corporation. [...] TCP/IP acts as an efficient pipeline, moving data from one point to another. [...] Almost **all the intelligence is on edge**, all the services, all the applications are created on the edge devices. Creating a new application does not involve changing the network. The Web, voice, video, and social media were all created as applications on edge without any need to modify the Internet protocol."*

However, from a market perspective, the current architecture creates unnecessary intermediaries in the flow of value. End-users cannot choose to buy the communication service directly from each provider in the route. They cannot even select a specific route or look for a certain level of service.

The Internet is a network only from a technological point of view, from the market perspective it is a chain of intermediaries. That is negative not only from the user's point of view but also from the service provider, as the communication service becomes a commodity with no way to provide better features at a premium price, even if the user needs it and wishes to pay for it.

Moreover, the settlement of payments between various actors is still a process requiring human intervention and trust between parties resulting in delays and frictions.

Every time a specific overlay network is conceived, it is to overcome the limits that IP has as a network protocol, so for the standard functionalities, it is assumed it has to provide.

So, while the intuition that applications (and therefore intelligence) should be at the edges must be preserved and strengthened, network-level technologies could be introduced to address the fact that current situation:

- does not provide the possibility to develop real distributed network topology that means a more efficient market,
- lacks a disintermediated market of communication services, and
- doesn't provide an acceptable level of privacy by design.
- creates the so called "The paradox of the best network"[4] that states that "the best network is the hardest one to make money running".

The main problem to adopt a new protocol stack that improves the situation at the network level is the tremendous difficulty to overcome the existing TCP-IP network effect. To push ISPs to buy new network devices implementing the new stack, the revisited Internet market model should

1. reward the ISPs that support the new stack and
2. provides a positive-sum for all stakeholders

This is the idea behind Nova.

2 PROBLEM DEFINITION

The Internetwork Protocol Version 4 was first described in IEN54[5] dated September 1978 and then standardized in RFC760[6] and RFC791[7]. It was a point of arrival of a long path that saw the creation of data networks in which Telecoms have been focusing on connecting devices, perpetuating the telephony model where devices and applications are the same.

The following technological limits of the stack design are symptomatic of this genesis and arising from its subsequent evolution.

2.1 Unfriendly roaming

Similarly to the way that the public switched telephone network switches calls between phone terminals, IP routes data between interfaces of devices. However, it is applications that wish to communicate, not the source and destination interfaces.

As the interface address and transport layer port number identify different applications, the network has not any such notion of "application." That is, the network uses information on where an application is located to determine which application this is. Every time connection point changes, from the network perspective there is a routing discontinuity managed by an upper layer, much-complicating multi-homing, mobility, and security.

Also this design concern the multi-homing, that is, two interfaces of the same host must have different addresses, so it had no way of knowing that they belong to the same application. This issue persists in IPv6 (see RFC4291[8]) as "IP addresses of all types are assigned to interfaces, not to nodes".

2.2 Poor layer separation

Also in 1978 TCP was split from IP. This could have been a good idea if the two layers:

1. were independent, and
2. didn't contain repeated functions.

None of both is the truth: to operate efficiently, IP needs to know what TCP is doing. The workaround of MTU discovery that TCP does to avoid IP fragmentation proves the point.

While, as early as in 1987 the networking community was aware[9] of the harmful of the IP fragmentation problems, it was not recognized as a symptom that TCP and IP were interdependent and therefore splitting it into two layers in that way was a bad decision.

2.3 Poor Quality of Service

Applications have poor ways of expressing their desired service characteristics to the network, other than setting a predetermined bit set to ask for the desired QoS, an option moreover not provided by socket API, or choosing a reliable (TCP) or unreliable (UDP) type of transport. There is no way to verify and enforce a Quality of Service to adhere to a Service Level Agreement.

There are no means for the sender to choose the most suitable route with a specific metric such as latency, jitter or bandwidth.

2.4 Mutation of layering

The original Internet architecture before January 1, 1983, when the NCP protocol was discontinued, was designed with three layers of increasing routing scope:

1. data link (to manage different types of physical media),
2. network (to manage different types of networks), and
3. internetwork (to manage a network of networks), each level with its kind of addresses.

The IP was therefore thought of as an internetwork layer protocol (over NCP as the network protocol) but was used as a network layer protocol. As a result, as well described by J.Day[10]

"the Internet ceased to be an Internet and became a concatenation of IP networks with an end-to-end transport layer on top."

The outcomes of this flaw are several: both inter-domain and intra-domain routing now happen within the network layer, and its scope has been artificially isolated by introducing earlier EGP[12] and later BGP[13] protocols.

BGP itself is flawed. Every BGP domain is operated by entities called Autonomous Systems (AS). The number of ASes multiplied in 20 years from 5.000 to 84.000 (see figure 1).

An AS trusts every other AS that packets are routed as advertised by the protocol. A core issue is BGP hijacking (RFC4593[14]) that is the illegitimate takeover of groups of IP addresses by corrupting Internet routing tables. Several solutions are proposed, none of these is satisfactory:

RPKI for BGPSEC or DNSSEC is documented

in RFC 6480[15]. The RPKI specifications spread out in a series of RFCs 6481-6493 in which, in a context of high political division, Internet players have to agree on a single root of trust to guarantee for everybody else, also creating a centralized organization again prone to political interference.

TLS PKI that relies on roots of trust and in which a certificate is accepted if a member PKI sign it. As this solution is secure as its weaker member, it offers little progress compared to the original situation.

Both solutions are the third level of complexity to solve an issue that a layer 3 ISO-OSI routing protocol should address natively.

2.5 Theory ignored

In 1981 Richard Watson provided a first theory[16] of reliable transport, which recognize that connection management requires only timers bounded by a small factor of the Maximum Packet Lifetime (MPL).

Watson and his team developed the Delta-t[17] protocol upon this theory, in which:

1. the sender and receiver can safely remove the state of a connection once the connection-state timers expire without the need for explicit removal messages, and
2. new connections are established without a specific handshake phase.

On the contrary, TCP uses both the explicit handshake along more limited timer-based management of the connection's state. Had TCP incorporated Watson's results it would be more efficient, robust and secure, eliminating the use of SYNs and FINs and therefore all the associated complexities and vulnerabilities to attack (such as SYN flood).

2.6 Bad congestion control

In 1986 the Internet experienced a congestion collapse. To improve the situation, hastily, it was decided to adopt in TCP the congestion avoidance scheme from Ethernet networks with a few modifications. It was an unfortunate decision because it maximizes the value of the congestion notification delay (i.e., reaction time) and its variance.

TCP implicit congestion avoidance mechanism is greedy: by definition, it needs to wait congestion happen to act and may be triggered when the network is not congested, causing a downgrade in performance.

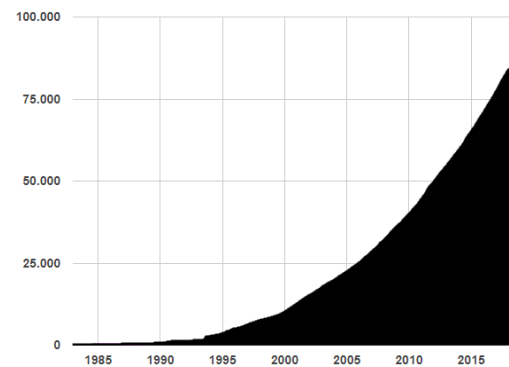


Figure 1. Number of Autonomous Systems. Source: Regional Internet Registries Statistics[11]

Furthermore, when congestion occurs, all endpoints are treated equally by the network; this "equality" is not economically justified and doesn't fit a world of different applications.

"the simple flat pricing structure that is traditionally associated with this sort of resource sharing doesn't provide any incentive for applications to release expensive resources that can be used by applications that need them more and are willing to pay for them. [...] Bandwidth is consumed by software running on devices rather than by humans, and there is no upper bound on the bandwidth an application may require. Applications are digital goods which cost almost zero to reproduce and distribute." [2]

2.7 The addressing scheme affects topology

In 1964 P. Baran, while working at RAND looking for surviving networks in the event of a nuclear exchange, wrote a renowned memorandum[18]. His analysis identifies three possible approaches, a centralized system, a decentralized one or a distributed one.

Baran's solution was a distributed network design in which packets could be rerouted among multiple potential redundant paths, and the loss of any single point in the network would not result in a critical fail.

From the network perspective, the problem that the military then saw as the survival of the command and control system becomes today the problem of having a topology resilient to the loss of routes and nodes, and so, that can handle the world economy.

All available maps of the Internet, based on IP addresses, resemble a topology more like a decentralized than a fully distributed network.

The explanation of this situation lays in the fact that the class-based IP addressing, while helpful in the beginning to lure the join of more networks to the Internet, now is an incentive to keep up of clusters.

Due to this approach, an IP router does stateful routing, maintain a routing table and performs a time-intensive lookup every time a packet is received to forward it to the next hop. Keeping the network state in the memory means to be vulnerable to memory exhaustion (i.e., attacks can be designed to exploit this limit).

Today, with the availability of new technologies at the level of linking (i.e., wireless, WiMAX) and with the emergence of IoT, class-based addressing reduces the number of interconnections and creates erroneous economic incentives that imply higher connectivity costs due to lower market competition.

Competition can be significantly influenced by the topology of the network and the ability of a selected number of players to control bottleneck resources in its parts.

The ideal would be to have technological assumptions in the design of the network that leave the nodes to choose how much they want to pay to be more interconnected and therefore how much the network should be distributed.

2.8 Lack of security

The TCP-IP protocols were born in a world that was entirely different from the hostile environment in which they currently operate.

Fundamentally, the Internet's building blocks are the same core protocols adopted by the ARPANET more than four decades ago, above which new ones were built to extend functionalities matching new requisites (i.e., IPsec, DNS, ...) or to fix lacks in the design (i.e., BGP, ...).

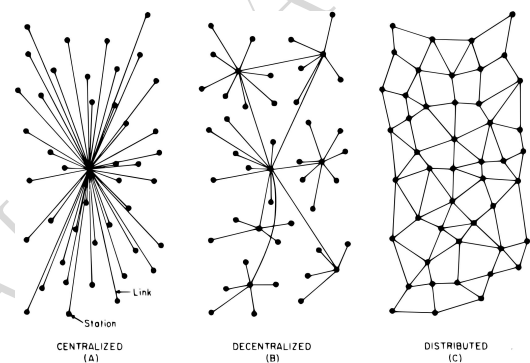


Figure 2. The three topologies alternatives (from Baran[18] memorandum)

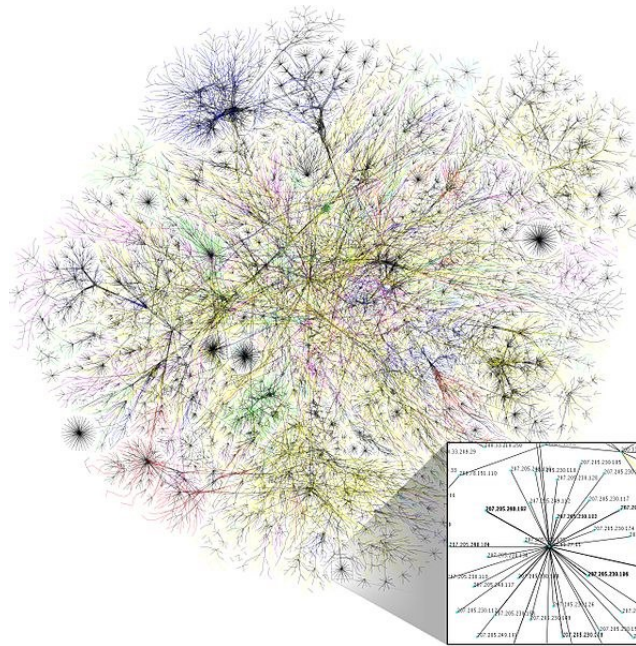


Figure 3. Partial map of the Internet based on the January 15, 2005 data found on opte.org. Each line is drawn between two nodes, representing two IP addresses. The length of the lines are indicative of the delay between those two nodes. This graph represents less than 30% of the Class C networks reachable by the data collection program in early 2005.

Although there is an extensive analysis[19] of the possible attack vectors to the TCP-IP stack, today we are aware that security is, first of all, a mindset thence a feature, that has developed extensively after the design of TCP-IP. As the complexity of the Internet protocols stacks grown, the more the attack surface increases.

Limiting the digression to the IP protocol only, it lacks some essential security feature that should be expected in a network in which an intermediate node can identify the endpoints such as:

1. packets shouldn't be hijacked and redirected without the sending party being aware of it, and
2. sending party must be able to avoid routes through an adversarial or untrusted network.

2.9 Lack of privacy

The design of TCP-IP stack haven't the privacy in mind, routes (including the sender and the receiver) followed by a packet can be tracked at each hop in the route. Also, the BGP protocol doesn't let the user to choose the route and doesn't allow the use of more than one simultaneously for the same destination in a given instant.

At the transport level, data is in clear-text, and its encryption is (eventually) delegated to an upper level of the stack.

205 3 OVERCOMING LIMITS OF TCP-IP

Numerous attempts and proposals have been made to solve problems highlighted, some specific for some issues (as, for example TLS[20]), others as a general redesign of the stack architecture.

210 This document will briefly evaluate those among them who provide ideas that then helped in the design of Nova. A survey[21] by Din, Yang and Den provides the reader with further highlights on other redesign proposals.

3.1 Add a set of security extensions (IPsec)

In July 1992, the IETF started to develop IPsec (as a part of IPv6, then back-ported) that is officially standardized in a series of Request for Comments (RFC-1825[22], RFC-1826[23] and 215 RFC-1827[24], published in 1995) addressing various components and extensions. IPsec provides security for transmission of information over unprotected networks. It acts at the network layer, protecting and authenticating IP packets between participating IPsec devices.

IPsec in tunnel mode, encrypt and authenticate the entire IP packet then encapsulate it into a new IP packet with a new IP header. It will be decrypted and eventually forwarded by the 220 receiver. Tunnel mode is used to create virtual private networks (VPN) for network-to-network communications (e.g., between routers to link sites) and host-to-network communications (e.g., remote user access).

3.2 Overlay Networks

225 Overlay networks are built on top of the TCP-IP, connecting nodes by virtual or logical circuits, each of which corresponds to a path, maybe through many physical connections in the underlying network.

These try to build a distributed topology on top of a decentralized one with the following trade-offs:

- 230 1. its nodes are mostly located in leafs of the graph representing the underlying network, on one-degree vertexes, where the bandwidth and latency characteristics are worse,
2. routing doesn't take full advantage of the topology of the underlying network,
3. its traffic can be identified, discriminated and possibly blocked.
4. the bootstrap of a node requires to trust a set of nodes with whom no relationship can attest their reputation.

235 We focus here on two - deployed - privacy enabling overlay networks, Tor and I2P.

The philosophical difference between the two is that I2P tries to move existing Internet services into the overlay network and provide service implementations within the framework whereas Tor enables anonymous access to external Internet services implemented and operated separately.

240 While Tor has hidden services and I2P has exit nodes, the canonical usage of Tor is proxying communication to the regular Internet, and the canonical usage of I2P is accessing integrated services.

In the proxy case, both can be subject to the Exit Node Eavesdropping attack[25].

245 Since each hop is routing-aware only about peers, this approach eliminates the ability to get data on sender and recipient by doing on path packet analysis.

I2P and Tor also differ in many technical details presented in the following table 1.

3.2.1 Tor

250 The theoretical debate about "Onion Routing" started in the mid-1990 with the purpose of protecting U.S. intelligence communications online. In 2004 Dingledine, Mathewson and Syverson publish the Tor white paper[27] in which they outline an application-level protocol that implements "Onion Routing" as following.

Tor	I2P
3-hop tunnels	user-configurable, randomized number of hops
bi-directional tunnels	uni-directional tunnels
guards, bandwidth-based peer selection	performance-based peer selection
7 directory servers with complete data	super-peer DHT (floodfill peers)
link and layered-encryption, but not (necessarily) end-to-end encryption	end-to-end, link and layered encryption
many exit nodes, few hidden services	one exit node, many integrated service
hidden services are external TCP servers	build-in servers for many services
implemented in C	implemented in Java
transport over TCP only	transport over TCP or UDP

Table 1. Key technical differences between Tor and I2P. (from Herrmann[26])

Tor bootstraps and discovers the topology of the network connecting to one of the trusted directory authority that agree with each other a "consensus" topology approved via a distributed voting system.

Any Tor client receives a "consensus" table that provides information about all active nodes in the network and their metadata.

These metadata is then evaluated to build a circuit (source based routing), a chain of 3 nodes used to connect the sender to the receiver where the (possibly hidden) service is hosted.

The sender encrypts the data, including the destination IP address of the next node, several times and sends them through a long-term virtual circuit spanning successive random select node. Each such node decrypts an encryption layer to reveal the next hop in the route and pass the remaining encrypted data to it. The receiver finally decrypts the innermost encryption layer and sends the original data to the destination without revealing or knowing the source IP address.

The first relay used in Tor circuits is called "entry guard relay," and it's a privileged position because it can see the real position in the network via IP's of Tor users. For this reasons, it's "expensive" to become a guard.

Still, an adversary can make some hypothesis by analyzing the network latency between involved nodes[28].

3.2.2 I2P

I2P network[29] follows an approach called "garlic routing" in which multiple messages are bundled together and sent via short-lived unidirectional encrypted tunnels, decreasing the number of samples that an attacker can use to mount an active attack.

I2P network is fully distributed and self-organizing. Routes are built by continuously profiling and ranking performance[30] of the underlying connections, rather than trusting claimed capacity. Nodes selection is not uniform but weighted by available bandwidth. Faster nodes have a higher probability of being picked than slower ones.

I2P stores all topology data in a modified Kademlia distributed hash table (DHT), called netDB. The bootstrap is performed by connecting to any node in the network. Also, I2P can be subject to possible information leak due to network latency analysis[26].

3.3 RINA

"Recursive Inter-Network Architecture" principles was delineated in John Day book[10] and focus on the fact that are applications that wish to communicate so computer networking is

just Inter-Process Communication or IPC.

It's an entirely rethought of the network stack with a model that comprises a single repeating layer, the DIF (Distributed IPC Facility), which is the minimal set of components required to allow distributed IPC between application processes.

RINA intrinsically supports mobility, multi-homing, and Quality of Service. RINA architecture separate protocols mechanism from policy dramatically reduces the number of protocols required in the architecture, while still allowing each layer to be customized to provide different levels of quality of service.

This complete redesign has trade-off as its scope is not limited to routing and transport protocols but offers a general solution regarding also the functions of higher level protocols.

The implication is that the change of the architecture involves not only Internet Providers but imposes, in perspective, a complete redesign of all the applications that use it. This broad scope implies that an eventual transition from TCP-IP to a network designed accordingly to RINA have far-reaching implications.

3.4 SCION

"Scalability, Control, and Isolation On Next-Generation Networks"[31] [32] was an inter-network protocol, a replacement of functionalities of BGP. SCION lies upon the concepts of Isolation Domains (ISD) where a set of Internet service providers agree on a single root of trust and upon SCION headers containing digitally signed paths for the packets inside and between the isolation domains.

ISD likely aggregate Autonomous Systems on a consistent legal environment basis (as suggested by SCION authors) so there is a risk of Internet balkanization due to political dynamics. While SCION ensures an absence of global kill switches (Unlike BGP, no entity can cause an outage of an ISD by operating outside the ISD), it still relies on centralized trusted parties (PKI) inside an ISD (local kill switch). A mitigation factor is that a provider can be a member of multiple ISDs.

To join an ISD (i.e., become a member of an ISD), an AS needs to be connected to it and needs to accept its regulations and policies. An ISD specifies accepted authorities, which are commissioned and authorized to provide digital identities and cryptographic keys for the entities inside the ISD.

A SCION address consists of an ISD identifier, an AS identifier, and an end-host address. As end-host addressing in SCION is local to an AS, it allows ASes to use different address spaces (e.g., IPv4, IPv6, or MAC addresses) to address their end hosts.

Consequently, SCION fixes the J.Day missed internetwork protocol issue as it permits hybrid addressing, that is, source and destination can have addresses of different types.

It also permits incremental deploying as an ISP needs only a single border router capable of encapsulating and decapsulating SCION traffic as it leaves, enters, or traverses its network.

Its key value propositions for ISP, as stated in SCION book[33], are:

1. higher global interconnection availability that can be embedded in SLA to customers,
2. finer control of traffic traversing their domain which can help with traffic engineering,
3. "guaranteed inter-domain bandwidth" service, and
4. reduced operation running costs.

Therefore SCION is a technological proposal that fits the current business model of the Internet provider. SCION permits the modeling of the current complexity while leaving room for different governance models of ISD.

3.5 HORNET and TARANET

HORNET[34] is a network-layer anonymity protocol and therefore can be seen as a replacement of the IP protocol itself. It is built to provide high-speed low-latency onion routing.

It offers relationship anonymity as defined by Pfitzmann and Köhntopp[35] that prevents linking the two endpoints.

HORNET doesn't require per-flow session state on intermediate routers. Instead, the session state is offloaded to the endpoint, who then embed this state into packet header built in a way that each intermediate node can later extract its state as part of the packet forwarding process.

To establish a route, it performs a setup phase in which HORNET uses two Sphinx[36] packets to traverse all nodes on both forward and backward directions and establishes a per-session state with every intermediate node.

Processing Sphinx packets involve expensive asymmetric cryptographic operations; thus Sphinx alone is suitable only for the setup phase in which the sender establishes a set of symmetric keys, one for each node on the route. Then, in the data transmission phase, HORNET use the symmetric key to check the header's integrity, onion-decrypt the data payload, and retrieve the information to route the packet.

HORNET operates through source-based routing, in which an endpoint knows the network topology and selects a route through that topology to reach the destination.

Unfortunately, source-based routing does not scale because the sender would need to know the entire Internet topology to discover routes.

The HORNET white paper does not provide a solution to this problem within the protocol itself and suggests the use of next-generation Internet architectures that use source-controlled routing (e.g., SCION (see 3.4), NIRA[37], or Pathlet[38]), or on the Internet through IPv6 Segment Routing[39].

HORNET defends against passive adversaries that match packets upon its content achieving bit-pattern unlinkability and route information onion encryption.

Nevertheless, HORNET is exposed to more sophisticated active attacks as all data packets in a session reuse packet headers and payloads are not integrity-protected. Consequently, HORNET doesn't shield against packet replays as an adversary could change the payload, creating a packet that looks like a legitimate packet to the next hop. This replay attack allows the insertion of recognizable patterns that can be used in combination with traffic analysis to facilitate the de-anonymize of communicating endpoints.

TARANET[40] is an evolution that solves some of HORNET stated shortcomings. It's a traffic-analysis-resistant anonymous protocol; in setup phase it uses mixing[41] to prevent linking messages based on their timing and order and in its data transmission phase it maintains the same constant transmission rate for outgoing packets belonging to a flow to resist traffic analysis.

TARANET also add payload integrity protection and replay protection.

4 GOALS OF THE NOVA STACK

The general purpose of the Nova stack is to transport encrypted payload from a sender application to a receiver one, through one or more routes, performing in real time the settlement of the payments necessary to operate it by the network.

370 Nova introduces the transport economic value in the network stack, taking note of the fact that we live in the Information Age, a period in human history characterized by the shift from traditional industry that the Industrial Revolution brought through industrialization, to an economy based on information technology.

In this age, the information and its economic value are the two sides of the same reality.
375 The insight contained in the design of this stack is that by reconnecting them together in the foundations of the new Internet, we can achieve an increase in efficiency and more transparent business models for everything that is built using the Internet (see section 4.1).

To operate the traffic payment system, we need a monetary unit of account that generates expectations of stability in relation to routing and transport prices, that is, a medium of exchange
380 embedded in the stack anchored to the value of the raw information transported (see section 4.2).

This choice implies radical consequences regarding security and privacy, which to date were only offered by the use of overlay network privacy but which, with Nova, become a required standard feature for all global traffic (see section 4.3) because a delivered payload is merged
385 with the corresponding economic transaction.

Nova's innovation is not so much in the technologies, as most of these already exist (at least in white papers) but in how these, coming from disparate fields, are assembled in a complete solution fundamentally different from any design so far.

4.1 Disintermediate The Internet

390 In the introduction, we said that, from the market perspective, the Internet is a chain of intermediaries. Each cluster of network hops along the value stream aggregates costs; an end user buys from his ISP a generic "access to internet" service without any warranty in the contract about network characteristic once packets go beyond the ISP network. The intermediate carrier i -th running a cluster of routers aggregates the prices for all subsequent j -th carriers. At best it
395 merely turns these prices at cost; otherwise, it can impose a top-up. As statutory warranties where not "inherited" along the route, intermediaries don't add any value to the service provided by another internet operator to the user, on the contrary, it's likely that user gets increased price while losing any transparency about the costs formation mechanism, losing any benefits that technology can provide to enhance competition.

400 We can observe that, from this opaque model, derives further opaque models for the business of the applications built upon it that pretends to be gratuitous, for example, ones based on the sharing of personal data as payment.

Nova instead makes the cost of each hop transparent to the end user, creating the ability to manage traffic with specific policies for each application used. It is a network in which the
405 end user can express its preferences and buy the bandwidth for which he is willing to pay; a *market managed network*[2] in which economic signals play a crucial role in the operation of such network.

Each sender pays the used bandwidth in real time and a new kind of Distributed Ledger technology record each of these transactions. The value transport occurs end to end at the
410 maximum speed allowed by the network infrastructure together with the traffic underlying that value, eliminating at the root the problem of scalability and topology in which incur other digital media of exchange currently built atop of TCP-IP.

When an ISP connects a new router to the network, it defines a policy about the costs of

routing and transport for the available routes. No other configuration is required. The new
415 interconnections become available to all nodes in the network. At the moment a user starts to
use it, the ISP starts to cash out. In the initial phase, therefore, the ISP obtain an increased
turnover due to the faster timing of payments. It is in the interest of ISPs therefore to incentive
the flow of Nova traffic than of legacy traffic.

To use a route, the sender starts a process of route discovery and SLA agreement; at the
420 end of which parties have defined a digital contract that can be used to execute the payments
of the end to end traffic when the payload flows.

Disintermediation eliminates intermediaries whose cost to service has become higher than the
value they provide. Nova provides the missed function to do it, nano-payment for connectivity
costs, and so removes delays and frictions created by the chain of intermediaries.

425 As consequence of the transparency that the vast amount of readily available connectivity
information creates, it empowers end-customer applications and makes it fast and convenient to
compare offerings by network characteristic and price—regardless of seller.

At the same time, pricing is used to control network provisioning and signalling.

430 *"Pricing can be viewed as a control mechanism for shaping demand. [...] By
increasing prices an operator can reduce demand, reduce congestion, and ensure that
services are provided to the users that benefit most and are most willing to pay. On
a short timescale, pricing can provide a type of flexible policing mechanism. On a
long timescale, it can be used as a part of a feedback loop to stabilize the network
through a sort of flow control. Viewed as a control mechanism, charging has the
435 advantage that it scales easily with the size of the network. [...] Price information
that is signalled to the edges of the network can play a significant role in providing
rational end-users and applications with the appropriate incentives to control their
flows." [2]*

Disintermediation decreases cost and allows the carriers to both increase margins.

440 4.2 A non deflationary medium of exchange

The Nova embedded unit of account, called "Moneta Nova", does not aim to establish itself
through the hype of rising prices due to absolute scarcity privileging the store of value money
function but instead wants to set itself as a medium of exchange for the network from the
beginning, focusing on stable prices and avoiding becoming a speculative asset. In practice, MN
445 adoption is successful if ISPs minimize the need to update prices within routers for monetary
reasons.

Several approaches have been proposed to achieve monetary stability of a decentralized and
censorship-resistant currency.

In Hayek Money, framed in Ametrano paper[42], price stability is achieved by dynamically
450 rebasing the outstanding amount of money: the number of units in every digital wallet is
adjusted instead of every single unit changing its value. Ametrano himself highlighted the main
problem of this approach:

455 *"the idea of a wallet changing its number of coins without any direct inflows or
outflows is surely an uncomfortable one. Currency volatility will not be discovered
by price volatility anymore but by the volatility of money amount instead: this is a
historical, cultural, and psychological change of paradigm hard to be accepted"*

Basis[43] outlines an algorithmic central bank in which a Three-Token System manages the
expansion and contraction of the monetary base. The "bank" auction off bond tokens when it
needs to contract Basis supply. Conversely share tokens ownership determines who is rewarded

460 in case of Basis supply expansion. Unfortunately both bond and share tokens fell under SEC regulation as securities, sinking the decentralization and censorship resistance completely. For this reason Basis team choose to return raised capital and shutting down the project[44].

Since Nova is a network stack formulated from scratch, it follows a different approach, a commodity exchange standard, and it embeds the latter in its technological foundation.

465 The overall adoption process envisaged is similar to that in the past has led gold to act as an anchor and to the birth of gold exchange standard. In the beginning, the value of gold cannot have introjected a nonexistent (at the time) monetary value. Initially, its utility for jewels arises its value; therefore it has acquired its price by its use as a commodity. Only later, for its optimal marketability and predictable supply and demand, it has become a medium of exchange, and
470 then a store of value.

We are looking for a digital commodity, and we like the one which underlies all of today's economy and society, the information itself. However, as information has a different value depending on each unit of content and for each economic agent, it isn't a commodity. We need a perspective that turns it into a commodity.

475 The only actors who can evaluate information as a commodity are those who do not need to know the content of the information and who provide a service based on the quantity of raw bytes.

In a network, these are the intermediate nodes under the constraint that information is encrypted (of which is unknown what it is) and anonymized (which is hiding the sender and
480 receiver), effectively commoditizing the information for price calculation.

If ISPs can hypothesize the content or the sender-receiver pair of the transmission, their pricing of routing and transport also incorporates the hypothesized value of content for the third party. So encryption and anonymization are mandatory in Nova not only for user privacy reason but also as a prerequisite for maintaining price stability.

485 Now that we have identified the commodity, to avoid to recur to bartering, we can build the MN unit of account representing a convenient multiple of the units of this commodity, i.e., encrypted bytes transported.

Practically the first early adopter defines arbitrary price in MN unit of account for the route and derives based on it the routing and transport prices of the next one. So it set a price for
490 MN in US dollar. Then the price of each new route is set based on previous prices. At the same time, a congruent monetary base is chosen in respect of the expected initial economic activity, that is the overall network traffic.

Summing up, at the bootstrap MN has a fixed exchange rate and an initial supply, and then the market is left free to price it versus fiat money.

495 Since the rules of monetary supply must be thought at the beginning once and forever to avoid the subversion of holder's expectations, it is a question of finding a model that generates expectations of price stability, which is to create a self-fulfilling prophecy. The idea is to keep the growth of the monetary base not exceeding the growth of the economy superintended by the medium of exchange.

500 Let's start from some considerations regarding the "Fisher equation of exchange" identity, which in its modern form is:

$$M \cdot V = \sum_i (p_i \cdot q_i) \quad (1)$$

where on the left, the supply of money that is the total value of money expenditures in all transaction:

M is the total amount of money available on average in an economy during the period,

505 V is the velocity of money, that is the average frequency across all transactions with which a unit of money is spent.

and on the right, the demand for money that is the total value of all items transacted:

p_i is the price of the i -th transaction,

q_i is the quantity of the i -th transaction.

510 Let's divide the monetary base into two sets, one that contains all the monetary units M_I that participate at least one transaction in the interval I , name "active money in I" and one that contains those M_O that are inactive in the same I .

If we separate the identity 1 into two identities for each set, only the identity for M_I still holds a meaning. In all economies, M_O is the money in the I interval that performs the function
515 of store of value for economic agents specifically as Keynes states in [45]:

(1) to bridge the gap between receipt of income and planned expenditures - the transactions motive;

(2) to provide a reservoir of purchasing power that can be used to finance unanticipated expenditures - the precautionary motive; and

520 *(3) to satisfy the desire to hold wealth in the most liquid form if one expects interest rates on alternative assets to rise, thereby causing capital losses - the speculative motive.*

Nova R-Stratum implements a C clock function which assigns a natural number $C(g)$ to any gram g , a Lamport timestamp[46]) that defines an irreflexive partial ordered set of grams. As an
525 economic transaction is a network message included in a gram, exists a **network-wide valid** interval I that encompass all transaction t starting from transaction a ending at transaction b , endpoints included.

$$I = \{ C(t) \in \mathbb{N} \mid C(a) \leq C(t) \leq C(b) \} \quad (2)$$

One of the two meters for the economic activity managed by the monetary unit is the velocity
530 of active money V_I that is the average frequency across all transactions with which a unit of money that participates to at least one transaction is spent. V_I is obtained by solving the identity 1 accordingly; the numerator is the sum of all confirmed transactions timestamped in the interval I , and the denominator are the unit of monetary base employed at least one time in these transactions M_I .

$$V_I = \frac{\sum_{n \in I} (T_n)}{M_I} \quad (3)$$

535 Due to the construction of the equation 3, the I minimum length is $[a, a]$ and represent a set of confirmed transactions about which we can't say which came first; it means that they couldn't have affected each other. As Nova have a construct to prevent double-spending, in this condition $V_I = 1$ by design. if we enlarge this interval, it becomes possible that the same monetary unit used for a transaction at timestamp a is reused for another transaction in a
540 timestamp t subsequent a . Therefore, by introducing the concept of active money, the active money velocity $V_I \geq 1$.

Unexpected changes in velocity V_I would thwart attainment of the goals of monetary stability, however, if the monetary consensus rules of Nova use only the growth in the money supply as a measure of the impact of monetary policy on the economy. Fortunately, this data is retrievable
545 and we can take this parameter into account when defining the rules for the issuing

The other meter for the economic activity is the activity ratio A_I defined as:

$$A_I = \frac{M_I}{M} \quad (4)$$

When the money demand for transactions is matched with a scarce offer (i.e. was higher than the monetary base), then to get it, the agents of the economic system would be willing to offer a larger amount of goods in exchange, or they would reduce good prices. That is a monetary deflation situation.

If the supply cap is fixed and predefined, the monetary deflation is the norm during the adoption of the monetary unit. The inelastic supply, that is, leads to hoarding for investment purposes until the technology adoption curve stabilizes, and consequently, the expectations change. Until that moment it is subject to extreme volatility.

"this is the cryptocurrency paradox: possibly the best ever kind of money by most metrics, marred by the severe inability to serve as reliable unit of account" [42]

L.H. White[47] tackles the subject of the difference between gold and Bitcoin showing that a more elastic supply mechanism, like the one provided by gold, can be preferable as *"the rise [of prices] will bring about a much larger increase [of the supply] in the longer run by incentivizing owners of gold mines to increase their output"*.

As the network knows for a given period both A_I and V_I , it can try to prevent situations of deflation from monetary base scarcity by increasing the latter's offer, shaping the expectations of economic agents towards the monetary stability, expectations that cryptocurrencies with a monetary base determined once and for all do not create. On the contrary, if the economic activity is being reduced, the network will cease to create new monetary base till activity returns above the point of maximum reached previously.

This fact is particularly important to avoid deflation during the adoption phase when the economy over-taken by Nova can grow significantly.

The newly minted monetary unit is distributed via a proof of bandwidth mechanism that rewards ISPs that provide additional bandwidth to the network, radically changing their business model and opening up a whole world of business opportunities for them.

It creates a further incentive for ISPs to push the transition from TCP-IP to Nova as all new traffic means greater economic activity overseen by the new network and therefore greater money generation reward for them.

Nova also offers the opportunity of operating other transactions involving goods and services, MN units of account will denote them.

As the network grows, the monetary unit will oversee a more complete and complex economy, and all these transactions will be taken into account for determining the overall economic activity of the network and the velocity of circulation of the money.

More details of monetary unit generation and consensus rules from a technical perspective can be found in 7.8.

4.3 Privacy by design

"Privacy by design" is an approach to confidentiality formalized in the last two decades (see [48] and [49]) and also adopted by the recent European Union General Data Protection Regulation (GDPR) of 2016.

The Nova network complies natively with all the fundamental principles[50] of "Privacy by design" from the moment the information is encapsulated before leaving the sender, to the one in which the receiver decapsulates it.

The stack is proactive, not reactive, in the functions it makes available, it anticipates and prevents privacy-invasive events such as:

- leaks of the information carried,
- the discovery of the type of information transported,
- the determination of the route beyond the directly adjacent nodes and,

- the identification of the node position in the network topology by third-party nodes.

Nova, **in the scope of functions expected by a network stack**, achieves an End-to-End Lifecycle Protection by providing "relationship anonymity" as defined by Pfizmann and Köhntopp[35] that prevents linking the two endpoints.

However, as any network-layer anonymity system, an adversarial first-hop can discern the sender while similarly, a compromised last-hop can distinguish the receiver. The requirements to have in the endpoint device at least a routing node and the internal virtual adapter is to mitigate as far as possible this unavoidable issue. Moreover unlike overlay networks, in most situations a node knows the first-hop it connects to as the user choose his ISP and a legal contract formalize the terms of service between the parties.

It is important to note that the trust in the peer required during the first connection does not imply the same during traffic payments.

Nova delivers Privacy as the Default; if an individual does nothing, their privacy remains unimpaired. No action is needed by the user to protect its privacy; it is built by default into the system, deeply embedded in the architecture of Nova, and it is not bolted on as an add-on or build upon flawed privacy designs, like in TCP-IP based solutions.

The result is that privacy is integral to the core functionality being delivered, without diminishing it. The use of End-to-End encryption makes it entirely useless for any intermediate part to long-term store the packets transported.

To achieve this level of privacy while maintaining the scalability of the solution, Nova disperses local onionized routing information in neighboring nodes and stores long-range ones in a distributed facilities system. A node would not need to know the entire Internet topology because it does not use the classic source-based routing, but a hybrid that still allows the sender to choose the routes to use.

4.4 Positive sum for stakeholders

The design of the protocol stack seeks to accommodate all legitimate interests in a positive-sum avoiding unnecessary trade-offs.

For example, the user's privacy obtained with the encryption of the routing fields and the transported payload also implies the fact that all those who run intermediate nodes are not able to be aware of the nature of the information being carried nor of their routing. That means that achieving more privacy for the user is obtained with fewer compliance costs for the ISP.

In economic sense it is a truly open technology; like IP, it is complementary to the underlying transport services. Unlike ATM in the mid-1990s, Nova don't mess with network cards and drivers

Finally, unlike in the IP addressing, the generation of a Nova address does not rely on an external set assigned by authorities. This result in more privacy and less burden for service providers.

4.5 Market friendly net neutrality

"Net Neutrality" is a commercial and political extension of the end-to-end principle, which holds that Internet Service Providers should not discriminate on the content, structure or profile of the traffic passing through it.

Nova defends the net neutrality while at the same time it preserves the ability for ISPs to modulate their business practices in an optimal way decoupling it from business actors that operates at an upper layer.

That is achieved by the combined facts that the sender chooses the routes to a destination and that the costs of transport and routing advertised by each intermediate node can be chosen by ISP only in respect of the required band characteristics and not in respect of the type of traffic as this is indistinguishable.

An ISP can define two logical routes with different performances and costs on the same physical route, whose choice is exclusively made by the user who can set its endpoint node policy to use cheaper routes for traffic such as email while using the most expensive ones for example for a video stream. With Nova, the user retains the choice. Not the ISP, which cannot carry out any traffic discrimination.

4.6 User centric

The end user is the only one who knows the value of the traffic generated and the characteristic he prefers for it. Traditional approaches to providing QoS let intermediate nodes in power to decide how to differentiate traffic. However, applications may benefit significantly from the ability to choose between offered service level agreements. In Nova, each application have a different traffic policies and the user can tune it. While for a video stream you are willing to pay for higher performance, an asynchronous transmission such as email can safely use routes with lower costs and performance. Rather than being forced to use a "one size fit for all" network service, Nova keeps it user-centric, focused on the individual choice, declinable for every single application and modifiable over time. This approach is in line with "end to end principle", in which the internal network nodes are kept simple, and complexity moved to the edges where information about user stays.

660 **5 OVERVIEW OF THE NOVA STACK**

The stack design is organized in 3 strict strata ("layers", see figure 4). "Strict" means that an underlying stratum doesn't use features of a higher one and that an interface defines calls between boundaries.

665 Unlike the classification proposed by RFC1122[51] for TCP-IP, each stratum have a specific Protocol Data Unit. This means that Nova has one encapsulation more than TCP-IP and that the entire Nova stack need only one EtherType value.

To avoid confusion with other protocols or stacks, we have chosen to use Latin names in Nova. For clarity these names (or the corresponding abbreviations) should be used.

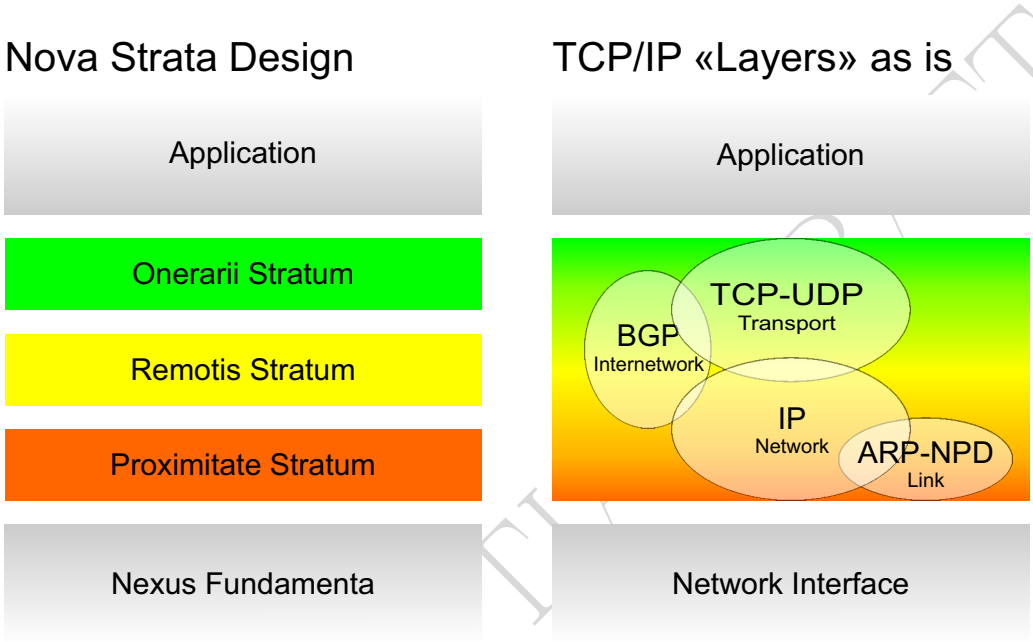


Figure 4. Nova Strata design compared to TCP-IP stack as is.

670 In Nova terminology, we refer to the heterogeneous technologies of the ISO-OSI layer 2 ("Data Link Layer") commonly implemented in the network card driver as "Nexus Fundamenta".

Above this, the lower level of the Nova stack is the "Proximate stratum" (**P-Stratum**, see section ??). It's goal in the architecture of the stack it to create a uniform graph as characteristics and topology for the upper layer. Practically, P-Stratum discovers one hop distant node, perform the handshake with them and establish encrypted connections with them while managing fragmentation issues.

675 The next level is the Remotis Stratum (**R-Stratum**, see section 7) contains all the features necessary to route data across the network from the sending node to the recipient node, making the related payments.

680 Onerarii Stratum (**O-Stratum**, see section 8) provides two protocols, a connection oriented TCP-like one that logically supports a permanent, reliable, flow controlled, full duplex, labeled bit stream connection between two nodes and a connectionless one, a minimal message-oriented like UDP.

For the sake of clarity, we will correctly use the following terms, contextualizing them for the stratum X referred to in the following paper sections. Data is :

- 685
- **submitted** when X get it by an overlying stratum to be processed,
 - **sent** when X submit processed data to an underlying stratum,
 - **received** when X get it from an underlying one, and
 - **delivered** when, after being processed by X, are returned to the overlying stratum.

690 A **Stratum Data Unit** (SDU) is information that is effectively transmitted as a single unit among peer entities of at a lower bound of a stratum. Instead, a **Process Logical Unit** (PLU) is how the information is yielded by a process inside a stratum to be aggregated or split in SDUs.

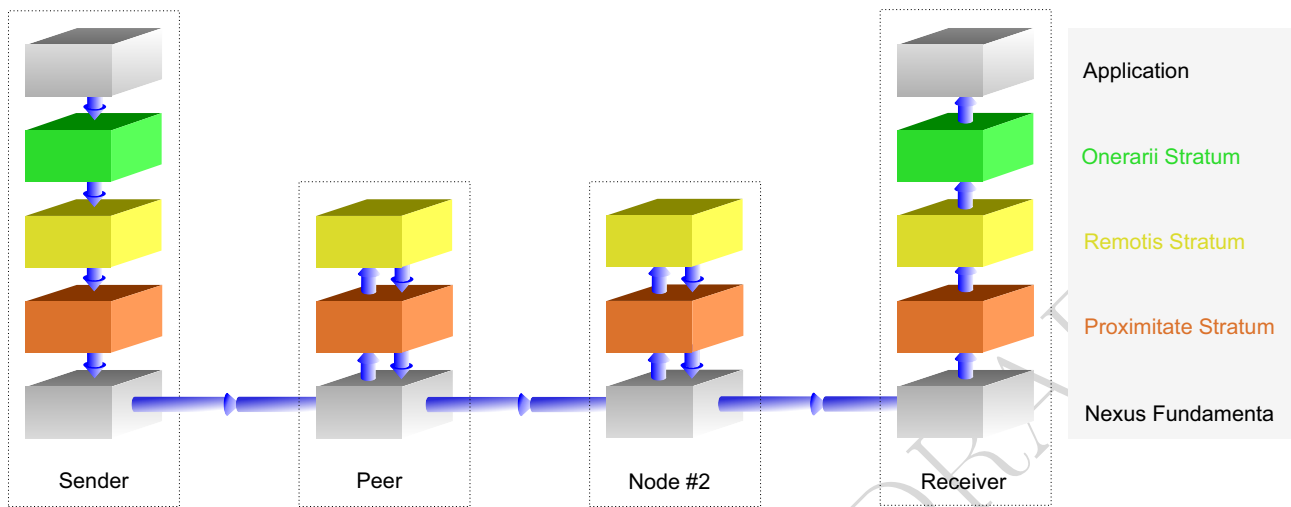


Figure 5. Application data flowing between strata and peers.

6 PROXIMITATE STRATUM

CONFIDENTIAL DRAFT

7 REMOTIS STRATUM

R-Stratum is the middle layer of the stack. Its purpose is to solve all the issues related to establishing a communication between two nodes at any distance between them. To achieve this result a series of processes are executed, specifically:

1. as source based routing can not scale to the size of the Internet because a sender would need to know the entire network topology to determine routes, it performs a **topology discovery** (see section 7.4).
2. the sender uses the information retrieved at step 1 to calculate possible routes. Not all routes are valid as the sender still doesn't know anything about information that could change unpredictably at any time like routing and transport prices at intermediate nodes. To acquire these data, it conducts an **link identify** (see section 7.5) by contacting intermediate nodes.
3. the sender finds what trails are compliant with its pricing/characteristic policies (PCP in the following) and then executes a **flow setup** (see section 7.6) by agreeing to the price and band characteristics advertised by routers. The receiver now is aware that a flow is probably needed in the opposite direction and can eventually start this same process to establish it.
4. the sender starts the **data transmission** (see section 7.7), acting as an endpoint of a transmission, paying in real time the costs involved. Intermediate nodes forward data not intended to themselves, that is to perform the function of a router. This also includes announcing the corresponding payment transaction to the network.

The R-Stratum **Core** performs the above processes on every node while **Facilities** run distributed processes necessary by the network to run properly.

1. The Beacon Facility (B-Facility, see section 7.4.2) gathers information on the network topology pro-actively.
2. The Validating Facility (V-Facility, see section 7.8) implement the issue of a new monetary unit with which the transaction validation and recording also takes place.

It is not necessary that all nodes run facilities; it is only necessary that there is a certain "density" of nodes running each of them within the network.

The functionality to interact with a node providing a facility's service is provided by the core.

7.1 Nodes of the network

A **node** is a distribution point of the traffic in the network. A node acts as a **router** if it receives data not intended to it and that it forward along the route to the actual destination. It provides a service to the network (i.e., other nodes). Any node in Nova can participate in the routing if it's willing to do so and is connected to at least two other nodes.

A node can also act as **endpoint** if originates or it is the destination of the traffic. Transmission occurs between two endpoints, a sending node denoted as the **sender** and a receiving node as the **receiver**. An endpoint transmits data originating from itself or receives data intended to him. It consumes a network service.

Each node embeds a set of long-term digital asymmetric key-pair generated locally, one for each cryptographic set supported by the node. Instead, a node rely on multiple sets of ephemeral key-pairs to encrypt and authenticate when performing routing.

A **node address** is obtained by concatenating a *varint* preamble that encodes the identifier of the cryptographic set used and the corresponding hash of the long term public key. A node has many node addresses as it can implement different cryptographic sets to be utilized (see section A.1); these addresses are concatenated in a **node address string**. The node address string is the information that another node must know in order to start a communication with it.

When it is used outside the stack (i.e., DNS records, QRCode) a checksummed base32 encoding like BECH32[52] SHOULD be used.

745 The term **node label** is used instead to indicate the hash of the ephemeral public key used for routing.

7.2 Nomenclature and notations

A **route** between the sender n^s and the receiver n^r is an unweighted and undirected sequence of nodes whose pairwise relationships connect the two nodes. Discovering routes is the result of the *topology discovery* process.

750 The **distance** d is the number of hops between two nodes n^s and n^r on route. A **neighbor** is a node within a defined distance d_{max} from n^s .

A **router** node n^i is the node at a distance i from node n^s on a route. A router receives data that is not intended to it and that it forward along the route to the actual destination.

755 A **close** is the node n^1 at a distance 1 from node n^s . An **edge** between two closes is a way through which two close can send unit of information.

Since there may be multiple routes from n^s to n^r through the same close with different distances, and a node may have multiple closes each on a different route to the other node, to disambiguate, a route is encoded as a tuple $(n^s, n^1, \dots, n^i, \dots, n^r)$.

760 The difference between the concept of R-Stratum edge and P-Stratum path is that two paths can exist between a pair of peers strictly on different adapters and this situation is not exposed to the upper stratum.

The degree of a node $|n|$ is the number of closes connected with n that is, the number of edges connected to it. A **leaf** node is a node with degree 1, typically an endpoint application running on a device.

765 A **trail** is a sequence of nodes between two nodes n^s and n^r on which to construct the flow of data. Unlike a route, a trail is weighted because each node advertise routing prices and upon an edge can be defined multiple **links** whose difference is that they advertise different band characteristics and transport prices to the other nodes of the network. Both trail and links have a direction because different prices and characteristics can be advertised in the opposite directions. A trail encoding add a link id j to link destination node $(n^s, n_1^1, \dots, n_j^i, \dots, n^r)$.
770 Discovering trails is the result of the *link identify* process.

A **flow** is a trail which is effectively set up for data transmission.

Sym	Meaning	Example
$\mathcal{R}$	a cryptographically secure pseudo-random number generator	$r \leftarrow \mathcal{R}()$
$\mathcal{G}$	a key generation algorithm that on input 1^k seed outputs public key/secret key pair	$(pk, sk) \leftarrow \mathcal{G}(1^n)$
$\mathcal{MG}$	a key generation algorithm that on input 1^k seed outputs master key pair and the master chain code	$(pk, sk, cc) \leftarrow \mathcal{MG}(1^n)$
$\mathcal{CKD}$	a child key derivation algorithm that on input parent extended key k^i , parent chaincode cc^i a child index j outputs derived key k_j^{i+1} and associated chaincode cc_j^{i+1}	$(k_j^{i+1}, cc_j^{i+1}) \leftarrow \mathcal{CKD}(k^i, cc^i, j)$
$\mathcal{E}$	an encryption algorithm that on input key k and message m outputs ciphertext c	$c \leftarrow \mathcal{E}(k, m)$
$\mathcal{D}$	a decryption algorithm that on input key k and ciphertext c outputs message m	$m \leftarrow \mathcal{D}(k, c)$
$\mathcal{S}$	a signing algorithm that on input key k and message m outputs a signature s	$s \leftarrow \mathcal{S}(k, m)$
$\mathcal{V}$	a signature verification algorithm that on input key k , message m and its signature s outputs true if the signature is verified	$(true \vee false) \leftarrow \mathcal{V}(k, m, s)$
$\mathcal{H}$	a cryptographically hashing algorithm that on input message m outputs its hash h	$h \leftarrow \mathcal{H}(m)$

Table 2. Algorithms notation

7.3 Grams

A **Gram** is the R-Stratum SDU.

We can then identify two sets of activities conducted by the R-Stratum:

- **Control plane** ones are the exchange of information necessary to enable the transfer of data submitted to a node and to run the network. Each piece of information is a **Message**, the control plane SLU.

As TARANET[40] setup phase, each node implements mixing before transmitting, in this case extended to all the messages of the control plane and not just the flow setup. A node does not send messages individually but queues them into batches of size m .

Once there are enough messages to form a **batch**, the sender first performs a Fisher-Yates shuffle of them and then transmits the batch as a single **gram**.

- **Data plane** ones that convey data submitted to the receiver. This means encapsulating data into a stream of flow grams, built as specified in TARANET data transmission phase, so that the intermediate nodes can forward to the receiver while a set payment fields, based on mumblewimble[53][54], permits the payment of traffic routing and transport.

Thanks to the underlying P-Stratum, which encrypts all traffic, it is impossible for an opponent who observes traffic on an edge between two honest nodes to distinguish chaff from actual data, control plane messages from data and different types of messages.

Chaffs defenses are used to defy traffic analysis attack aimed to identify communicating endpoints based on metadata such as volume, traffic patterns, and timing by adding padding traffic.

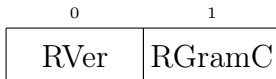
In the control plane, this is obtained by injecting chaff messages that implement the padding of the batches in the mixing process; a pair of close coordinates the injection to maintain a constant sending rate on a edge. The rate is calculated by probing underlying stratum performances.

Instead, data plane chaff and packet splitting achieve end-to-end padding so that a compromised node cannot distinguish chaff traffic from real traffic, and is thus unable to discover the real traffic patterns. The end-to-end constant-rate transmission is obtained by recovering necessities information during the link identify process, accordingly to prices advertised for each link and policies for the submitted traffic.

Sending or receiving a batch gram does not imply any payment for the parties involved, unlike sending a flow gram. In other words, a sender pays for the traffic actually submitted to the R-Stratum and not for control plane traffic. This difference implies that control plane messages specification **MUST** be defined strictly to avoid the abuse of submissions of hidden data.

7.3.1 Gram common header

The common header is used to check stratum versions and to differentiate between flow gram and batch gram processing.



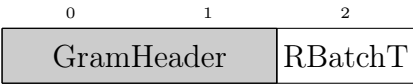
This header contains 2 fields:

RVer (1 byte) Each sent packet starts with a single leading byte, which indicates the version of the R-Stratum protocol used. A version of 0x00 means that no change is necessary, while a 0x00 version suggests that the node has deviated from the protocol specified initially within this document. A node **MUST** ignore grams with an unknown version.

RGramC (1 bytes) Gram class is set to 0x00 for control plane batch grams and to 0xFF for data plane flow grams. A node **MUST** ignore grams with an undefined class for the given

version.

820 7.3.2 Batch header



Message Types	Message Types name	See section
32	NEIGHBOR-FLOOD	7.4
33	NEIGHBOR-ACK-L	7.4
34	NEIGHBOR-ACK-R	
128	SPHINX	

Table 3. PDU Types

7.4 The topology discovery process

In the topology discovery process, a sender registers in the network discovering the local topology and looks for a sequence of labels to be used later in the header of a message to describe an anonymized route up to the receiver.

The basic idea is to have a distributed system that carried out the process in three phases. The challenge is to preserve privacy; it is not desirable to announce exact node position to the whole network, nor scalable that any node collects information about all nodes of the network.

The first one (7.4.1) operates in the neighborhood of a node and keeps anonymized information about routes between a node and a **gateway** in a distributed and redundant form. A gateway is a node that, when presented with the hash of the address of a destination node, can forward it through an anonymous route to that node. It starts when P-Stratum of the node n^s notifies R-Stratum that a new edge is available and is repeated periodically; the sender reaches out gateways (and beacons) along the new branch of the network reachable through that edge. At the end of this process, The second phase (7.4.2) gateways registers in the beacons known to them a label to forward the traffic to the sender. As the process takes place symmetrically in opposite directions on both closes of an edge when it is established, the sender can choose to become a gateway for other nodes.

The third phase (7.4.3) is executed when the sender have to retrieve topological data from beacon to discover long range routes to a specific destination. Dedicated nodes called **beacons** maintains pointers to gateways and associated labels to enable the discovery of a sequence of labels for long-range routing. A beacon run the B-Facility which keeps a shard of the global table pointing to gateways.

Sym	Meaning
$ n $	the degree of node n
$\langle n \rangle$	the obfuscated degree of node n (see section ??)
j	an iterator on a degree
k	an iterator on an obfuscated degree
z	a distance iterator
k, k^i, k_k^{i+2}	a key, the same of the i node on the route, the same for the node n^{i+2} reached along k -th edge from n^{i+1}
$\overrightarrow{k^{s,i}}, \overrightarrow{k_k^{s,i+2}}$	a key of the node n^s derived via CKD for the node n^i , the same derived for the node n^{i+2} reached along k -th edge from n^{i+1}
pk, sk	a public key, a secret key
cc, cc^i, cc_k^{i+2}	the chain code, used by CKD to generate the next depth level of keys, the same of the i node on the route, the same for the node n^{i+2} reached along k -th edge from n^{i+1}
w, w_k^{i+2}	a random number used as index to generate keys in CKD , the same for the node n^{i+2} reached along k -th edge from n^{i+1}
ed, ed_k^{i+2}	the corresponding w encrypted
p^i	an unencrypted payload generated at node n^i
e^i	an encrypted payload generated at node n^i
s^i	a signature generated at node n^i
L, lpk, lsk, led	the left label and the left pk sk and ed.
$R, rpik, rsk, red$	the right label and the right pk sk and ed.

Table 4. Symbols in the 7.4 section

7.4.1 Flooding neighborhood to find Beacon and Gateways

As the sender n^s knows nothing about the topology reachable through the new edge, as soon as the CLOSEADDED event is raised, it floods its neighbor nodes along the new edge till d_{max} with NEIGHBOR-FLOOD messages. Our thoughts on how to build a scheme that guarantees privacy starts from the flood phase of paper [55]. However, there are profound differences between discovery relationship paths in a decentralized social network, where the two counterparts exchange information off-network to identify each other, and conversely routes discovery in a communication network where flooding must discover who are gateways and beacons in the process. For this reason we change radically the flooding approach and move away from using private intersection set in path discovery phase, relaying instead on round-trips of messages with discovered gateways and beacons.

The first difference is that Internet topology is far less distributed than a social network one; nodes have less degree while routes are typically longer than social relationship paths. Pending further analysis, R-Stratum d_{max} is likely to be chosen greater than the one provided by the paper's extended scheme; therefore that schema must be addressed to become in some way "recursive", able to work on the paper for any d_{max} .

Secondly, we move away from source based routing by distributing the routing information in intermediates nodes. When a NEIGHBOR-FLOOD message is received by a node, it computes a route label L^i , that is the hash of an ephemeral public key generated at n^i ; this label is forwarded to all n^{i+1} in the flooding process. By repeating this subsequently on each node of the route, i -th node can address a message to head back to the sender n^s . For this reason we call it the **left** label of the route at i -th node. This label is kept in an entry of the routing table **RouteT** of a node (see table 5); if the OutLabel is set to 0 then any incoming message headed to L is arrived back at the original sender.

Field	NEIGHBOR-FLOOD	NEIGHBOR-ACK-L	Description
Node	n^{i-1}	n^{i-1}	The edge on which message come from
InLabel	L^i	R^{i+1}	The label of the route in the incoming message
OutLabel	L^{i-1}	R^i	The label of route in the outgoing message
OriginPK	$pk^{s,i}$	$pk^{r,i}$	The child public key of message originator derived to be used on i^{th} node
Edge	led^i	red^i	A unique identifier of the outgoing edge
Created	$T_{created}$	$T_{created}$	The creation timestamp of the entry

Table 5. RouteT table fields on node n^i

The same process of label creation happens when the message NEIGHBOR-ACK-L come back from the receiver to the sender. A label R^i is created at each hop, called the **right** label of the route at i -th node. The sender is now able to send a message to a replying node.

Once the NEIGHBOR-FLOOD packet has been received by a gateway or a beacon, on each intermediate node, have been defined all L labels necessary for a receiver to send back a message to the sender. Similarly, when the sender get back the NEIGHBOR-ACK-L message, on each intermediate node have been defined all R labels required for a sender to send a message to the receiver .

Thirdly, the sender and the recipient must exchange public keys so that, once a node has declared its identity at the beginning of the session, the exchange of information is encrypted and authenticated. It is not possible for the intermediates nodes to retransmit the same sender key because messages along that route discovery would have a common bit pattern, breaking bitwise unlinkability. To solve it, at each hop we should generate a child public key $lpk^{s,i}$ from

the sender key lpk^s according to the deterministic schema provided by the Bitcoin Improvement proposal 32[56] using the HMAC-SHA512 function as specified in RFC4231[57]. The schema can eventually be adapted for Curve25519[58]. All keys are non-hardened for the sake of this document.

885 To still apply the randomization technique described in the original paper, the node n^i have to know how many messages are needed by n^{i+1} to be sent to n^{i+2} . The retrieving of this information from the close n^{i+1} is performed by the underlying P-Stratum and exposed in the CLOSEADDED event as the obfuscated degree (**ODegree**), hereafter with the notation $\langle n^{i+1} \rangle$. P-Stratum detects and doesn't expose mutual closes that leads to $d \leq 2$ cycles and add
890 a positive noise to hide the network topology (see section ??).

Hence, the node n^i creates a set of tuple $lpk^{s \rightarrow i+2}, lcc^{i+2}$ of cardinality $\langle n_{i+1} \rangle$, using a random number as child index for the public key derivation function $\mathcal{CKD}_p$. The node n^{i+1} performs a shuffle on this set before the forwarding, resulting in further hiding the following topology.

Furthermore the message from n^i to n^{i+1} doesn't include lcc^{i+1} so it's impossible for n^{i+1} to
895 generate keys for n^{i+2} .

Field	Value	Description
Label	L^i	The label of a route at the given node
SK	sk^i	The private key of n^i
PK	pk^i	The public key of n^i
Created	$T_{created}$	The creation timestamp of the entry

Table 6. NodeT table fields on node n^i

Algorithm 1 describe how the elements necessary to build the starting NEIGHBOR-FLOOD message are computed on node n^s when the event CLOSEADDED is raised.

The R-Stratum of the originating node performs the master key generation function $\mathcal{MG}()$ to generate the master key pair pk^s and sk^s and the master chain code cc^s .

900 With these it can use $\mathcal{CKD}_p(lp k^s, lcc^s, j)$ function to computes the child extended public key $lp k^{s,1}$ and the child chaincode cc^1 from the parent extended public key $lp k^s$ and parent lcc^s for the new edge j .

R-Stratum also generates the label L^s by hashing $\mathcal{H}(lp k^s)$ which becomes the route local label on that edge for the sake of routing back any message to n^s from n^1 .

Algorithm 1 On sender node n^s , start NEIGHBOR-FLOOD direct propagation.

Require: $\langle n^1 \rangle$, the ODegree of the next node

```

1: on CLOSEADDED( $n^1, \langle n^1 \rangle$ ) do
2:   if  $\langle n^1 \rangle > 0$  then
3:      $(lp k^s, lsk^s, lcc^s) \leftarrow \mathcal{MG}()$ 
4:      $d \leftarrow 1$ 
5:      $w^1 \leftarrow \mathcal{R}()$ 
6:      $(lp k^{s,1}, lcc^1) \leftarrow \mathcal{CKD}_p(lp k^s, lcc^s, w^1)$ 
7:      $led^1 \leftarrow \mathcal{E}(lp k^s, w^1)$ 
8:      $L^s \leftarrow \mathcal{H}(lp k^s)$ 
9:     Insert in RouteT  $[0, L^s, 0, 0, 0, T_{now}]$ 
10:    Insert in NodeT  $[L^s, lsk^s, lcc^s, T_{now}]$ 
11:    for  $k \leftarrow 1, \langle n^1 \rangle$  do
12:       $w_k^2 \leftarrow \mathcal{R}()$ 
13:       $(lp k_k^{s,2}, lcc_k^2) \leftarrow \mathcal{CKD}_p(lp k^{s,1}, lcc^1, w_k^2)$ 
14:       $led_k^2 \leftarrow \mathcal{E}(lp k^s, w_k^2)$ 
15:    end for
16:     $p^s \leftarrow (d \| lp k^s \| lp k^{s,1} \| led^1 \| (lp k_1 \| \dots \| lp k_{\langle n^1 \rangle})^{s,2} \| (lcc_1 \| \dots \| lcc_{\langle n^1 \rangle})^2 \| (led_1 \| \dots \| led_{\langle n^1 \rangle})^2)$ 
17:     $s^s \leftarrow \mathcal{S}(lsk^s, \mathcal{H}(p^s))$ 
18:    Send to  $n_1$ :  $(p^s, s^s)$ 
19:  end if
20: end on

```

905 Algorithm 2 describes what happens on any node n^i upon the reception of the message from the close n^{i-1} . The message contains the label L^{i-1} , the public key $lpk^{s \rightarrow i}$ of n^s as seen by n^i , and a set of tuples of public key and chaincode, each one to be sent by n^{i+1} with a message to a close except the originating one. The set is shuffled before the forwarding so the previous generating hop n^{i-1} cannot do any assumptions on which key go to each node.

910 Similarly to what we saw at the sender, the intermediate node calculate the label L^i such any message that contain it incoming from node n^{i+1} can be forwarded to n^{i-1} headed to L^{i-1} . For each edge, except the originator, the process forward the flood, generating the public keys $lpk^{s \rightarrow i+2}$ and inserting an ephemeral entry $(L^i, L^{i-1}, L^{i+1}, lpk^{s \rightarrow i}, keyseed^i, T_{eph})$.

915 T_{eph} is a short-term deadline after which the entry will be removed from the table. The life of the entry will be extended if an NEIGHBOR-ACK-L eventually come back along the route.

The node creates and includes in each message also a set of child public key of the sender $lpk^{s \rightarrow i+2}$ and related chaincodes lcc^{i+2} that the node n^{i+1} have to provide to each of his n^{i+2} close.

Algorithm 2 On intermediate node n^i , NEIGHBOR-FLOOD direct propagation.

Require: $\langle n_j^{i+1} \rangle$, the ODegree of the next node along the edge of index j

```

1: on reception from  $n^{i-1} : (p^{i-1}, s^{i-1})$  do
2:    $p^{i-1} \rightarrow (d, lpk^{i-1}, lpk^{s,i}, led^i, (lpk_1, \dots, lpk_{\langle n_j^i \rangle})^{\overrightarrow{s,i+1}}, (lcc_1, \dots, lcc_{\langle n_j^i \rangle})^{i+1}, (led_1, \dots, led_{\langle n_j^i \rangle})^{i+1})$ 
3:   if  $\mathcal{V}(lpk^{i-1}, p^{i-1}, s^{i-1})$  then
4:      $L^{i-1} \leftarrow \mathcal{H}(lpk^{i-1})$ 
5:     if  $d = d_{max}$  or  $\langle n^i \rangle = 0$  then
6:       Insert in RouteT  $[n^{i-1}, 0, L^{i-1}, lpk^{s,i}, led^i, T_{now}]$ 
7:     else if  $d < d_{max}$  then
8:        $(lpk^i, lsk^i) \leftarrow \mathcal{G}()$ 
9:        $L^i \leftarrow \mathcal{H}(lpk^i)$ 
10:      Insert in RouteT  $[n^{i-1}, L^i, L^{i-1}, lpk^{s,i}, led^i, T_{now}]$ 
11:      Insert in NodeT  $[L^i, lsk^i, 0, T_{now}]$ 
12:      Shuffle  $\{(lpk, lcc, led)_1, \dots, (lpk, lcc, led)_{\langle n^i \rangle}\}^{\overrightarrow{s,i+1}}$ 
13:      for  $j \leftarrow (1 \dots |n^i|), n_j^i \neq n^{i-1}$  do
14:         $p_j^i \leftarrow (d + 1 \parallel lpk^i \parallel lpk_j^{s,i+1} \parallel led_j^{i+1})$ 
15:        if  $\langle n_j^{i+1} \rangle > 0$  and  $d < (d_{max} - 1)$  then
16:          for  $k \leftarrow 1, \langle n_j^{i+1} \rangle$  do
17:             $w_k^{i+2} \leftarrow \mathcal{R}()$ 
18:             $(lpk_k^{s,i+2}, lcc_k^{i+2}) \leftarrow \mathcal{CKD}_p(lp_k^{s,i+1}, lcc^{si+1}, w_k^{i+2})$ 
19:             $led_k^{i+2} \leftarrow \mathcal{E}(lp_k^{s,i}, w_k^{i+2})$ 
20:          end for
21:           $p_j^i \leftarrow p_j^i \parallel (led_1 \parallel \dots \parallel led_{\langle n_j^{i+1} \rangle})^{i+2} \parallel (lpk_1 \parallel \dots \parallel lpk_{\langle n_j^{i+1} \rangle})^{\overrightarrow{s,i+2}} \parallel (lcc_1 \parallel \dots \parallel lcc_{\langle n_j^{i+1} \rangle})^{i+2}$ 
22:        end if
23:         $s_j^i \leftarrow \mathcal{S}(lsk^i, \mathcal{H}(p_j^i))$ 
24:        Send to  $n_j^i$ :  $(p_j^i, s_j^i)$ 
25:      end for
26:      if  $(n^i \text{ is Gateway}) \vee (n^i \text{ is Beacon})$  then
27:        raise SENDANSWER( $L^{i-1}$ )
28:      end if
29:    end if
30:  end if
31: end on

```

Algorithm 3 On receiver node n^r , start of NEIGHBOR-ACK-L reverse propagation.

```

1: procedure SENDANSWER( $L^{r-1}$ )
2:    $(n^{r-1}, lpk^{\overrightarrow{s},r}, led^r) \leftarrow$  for RouteT.OLabel= $L^{r-1}$ 
3:    $(rpkr, rsk^r, rcc^r) \leftarrow \mathcal{MG}()$ 
4:    $w^{r-1} \leftarrow \mathcal{R}()$ 
5:    $(rpkr^{\overrightarrow{r-1}}, rcc^{r-1}) \leftarrow \mathcal{CKD}_p(rpkr, rcc^r, w^{r-1})$ 
6:    $red^{r-1} \leftarrow \mathcal{E}(rpkr, w^{r-1})$ 
7:    $R^r \leftarrow \mathcal{H}(rpkr)$ 
8:   Insert in RouteT  $[0, R^r, 0, 0, 0, T_{now}]$ 
9:   Insert in NodeT  $[R^r, rsk^r, rcc^r, T_{now}]$ 
10:   $w^{r-2} \leftarrow \mathcal{R}()$ 
11:   $(rpkr^{\overrightarrow{r-2}}, rcc^{r-2}) \leftarrow \mathcal{CKD}_p(rpkr^{\overrightarrow{r-1}}, rcc^{r-1}, w^{r-2})$ 
12:   $red^{r-2} \leftarrow \mathcal{E}(rpkr, w^{r-1})$ 
13:   $p^r \leftarrow (L^{r-1} \parallel d \parallel rpkr \parallel rpkr^{\overrightarrow{r-1}} \parallel red^{r-1} \parallel rpkr^{\overrightarrow{r-2}} \parallel rcc^{r-2} \parallel red^{r-2})$ 
14:   $e^r \leftarrow \mathcal{E}(lpk^{\overrightarrow{s},r}, ("Rcver" \parallel d \parallel \mathcal{H}(n^r) \parallel rpkr))$ 
15:   $s^r \leftarrow \mathcal{S}(rsk^r, \mathcal{H}(e^r))$ 
16:  Send to  $n^{r-1}$ :  $(p^r, e^r, s^r)$ 
17: end procedure

```

920 The relay back of NEIGHBOR-ACK-L is described in algorithm 4 the message is conveyed back to the originating node n^s .

The key $lpk^{s \rightarrow i}$ is used to onion encrypting information contained in a route-back message NEIGHBOR-ACK-L so that the ODegree of n^i required by n^s to efficiently build the route is not disclosed to any intermediate node.

Algorithm 4 Intermediate node n^i , reverse propagation of NEIGHBOR-ACK-L.

```

1: on reception from  $n^{i+1} : (p^{i+1}, e^{i+1}, s^{i+1})$  do
2:    $p^{i+1} \rightarrow (L^i, d, rpki^{i+1}, rpkr,i, red^i, rpki,i-1, rcc^{i-1}), red^{i-1}$ 
3:   if  $\mathcal{V}(rpki^{i+1}, e^{i+1}, s^{i+1})$  then
4:      $d \leftarrow d - 1$ 
5:     if  $d \leq d_{max}$  then
6:        $(L^{i-1}) \leftarrow \text{for RouteT.InLabel} = L^i$ 
7:        $(led^i) \leftarrow \text{for NodeT.Label} = L^i$ 
8:       if  $\exists L^{i-1}$  then
9:          $R^{i+1} = \mathcal{H}(rpki^{i+1})$ 
10:        if  $L^{i-1} \neq 0$  then
11:           $(rpki^i, rski^i) \leftarrow \mathcal{G}()$ 
12:           $R^i \leftarrow \mathcal{H}(rpki^i)$ 
13:           $w^{i-2} \leftarrow \mathcal{R}()$ 
14:           $(rpkr,i-1, rcc^{i-1}) \leftarrow \mathcal{CKD}_p(rpki^i, rcc^i, w^{i-2})$ 
15:           $red^{i-2} \leftarrow \mathcal{E}(rpkr,i, w^{i-2})$ 
16:          Insert in RouteT  $[n^{i+1}, R^i, R^{i+1}, rpki,i, red^i, T_{now}]$ 
17:          Insert in NodeT  $[R^i, rpki^i, rski^i, 0, T_{now}]$ 
18:           $(lpk^{s,i}, led^i) \leftarrow \text{for NodeT.Label} = L^i$ 
19:           $p^i \leftarrow (L^{i-1} \parallel d \parallel rpki^i \parallel rpkr,i-1 \parallel red^{i-1} \parallel rpki,i-2 \parallel rcc^{i-2} \parallel red^{i-2})$ 
20:           $e^i \leftarrow \mathcal{E}(lpk^{s,i}, ("INode" \parallel d \parallel lpki^i \parallel led^i \parallel e^{i+1}))$ 
21:           $s^i \leftarrow \mathcal{S}(rski^i, \mathcal{H}(e^i))$ 
22:          Send to  $n^{i-1} : (p^i, e^i, s^i)$ 
23:        else if  $d = 0$  and  $L^{i-1} = 0$  then
24:          Insert in RouteT  $[n^{i+1}, 0, R^{i+1}, rpkr,i, red^i, T_{now}]$ 
25:          if  $\text{PROCESSANSWER}(L^i, e^{i+1}) = \text{success}$  then
26:            SENDCONFIRMATION( $R^{i+1}, \mathcal{E}(rpkr,i, ("Rcver" \parallel d \parallel \mathcal{H}(n^s) \parallel lpk^s))$ )
27:          end if
28:        end if
29:      end if
30:    end if
31:  end if
32: end on

```

In sync with receiving NEIGHBOR-ACK-L messages, the sender populates a table of possible
925 routes (see algorithm 5).

Algorithm 5 Sender s , acknowledge reception processing.

```

1: function PROCESSANSWER( $L^s, e^1$ )
2:    $(lsk^s, lcc^s) \leftarrow \text{for NodeT.Label} = L^s$ 
3:   for  $z \leftarrow 1, d_{max}$  do
4:      $(pre^z, d_{exp}^z, lpk^z, p[\dots]) \leftarrow \mathcal{D}(lsk^{\overrightarrow{s, z-1}}, e^z)$ 
5:     if  $d_{exp}^z \neq z$  then
6:       return failure
7:     end if
8:     if  $pre^z = \text{"Rcver"}$  then
9:        $p[\dots] \rightarrow n^z$ 
10:      Insert in NeighborT  $[n^z, R^1, \{((lsk^{\overrightarrow{s, 1}}, lpk^1, led^1), \dots, (lsk^{\overrightarrow{s, z}}, lpk^z, led^z)\}, T_{now}]$ 
11:      return success
12:     else if  $pre^z = \text{"INode"}$  then
13:        $p[\dots] \rightarrow (led^z, e^z)$ 
14:        $w^z \leftarrow \mathcal{D}(lsk^{\overrightarrow{s, z}}, led^z)$ 
15:        $(lsk^{\overrightarrow{s, z}}, lcc^z) \leftarrow \mathcal{CKD}_s(lsk^{\overrightarrow{s, z-1}}, lcc^{z-1}, w^z)$ 
16:     else
17:       return failure
18:     end if
19:   end for
20: end function

```

The last step provided in algorithm 4 is the sending of an end to end encrypted NEIGHBOR-ACK-R message so receiver is aware that the NEIGHBOR-ACK-L is effectively received by sender.

From the moment n^r receives NEIGHBOR-ACK-R, that contains n^s node address and public
930 key, it is therefore possible to establish a bidirectional and encrypted communication with a gateway or a beacon. By doing ECDH, we can use symmetric encryption for better performance.

7.4.2 Registering a route

The last step of this phase consists in registering gateways in beacons. Each gateway
935 discovered, when receives NEIGHBOR-ACK-R, sends to every known beacon a GATEWAY-REGISTER message consisting in the concealed address of the other end of the route n^s and in the signature of the n^s plain address done by the same node.

To go further in the analysis of following steps, we brief about the facilities involved in the process.

Beacon nodes run B-Facility; its purpose is to run a process to gather information on the
940 neighborhood topology proactively and to keep data updated in the **authority** nodes.

A B-Facility maintains a set of tables:

- a gateways table (**GTable**) that contains a list of gateways to a given nodes present in its neighbor,
- an authorities table (**ATable**) of authority nodes and their authoritative address spaces.

945 When a beacon receives the GATEWAY-REGISTER message, it looks up in the ATable for authority nodes that are authoritative for that address. Authoritative means that authority has previously advertised the network via an AUTHORITY-REGISTER message that it holds a list of gateway with hash of the address matches the pattern field contained in the message

through a bitwise AND-masking. In this way, each authority is configured to be authoritative of
950 a subset of node address namespace. The **masking depth**, namely the number of bits not set
(0) in the mask, defines the memory and computational resources needed by the authority.

The beacon send to each authority, that is authoritative for the hash of the gateway address,
a GATEWAY-ENLIST message.

An authority maintains a set of tables:

- 955 • a gateways table (**GTable**) that contains a list of gateways to a given nodes present in its
neighbor,
- an authorities table (**ATable**) of authority nodes and their authoritative address spaces.

7.4.3 Resolving an address in a sequence of intermediate nodes

- 960 1. notifying the authoritative TD-Nodes known when the address of a node that respects
this form is registered with a TD-Node.
2. making an authoritative TD-Node notify the other known authoritative TD-Nodes when
a mask-compliant address is registered.

7.5 The link identify process

By adding this information, valid routes are identified and involved edges of the graph get direction and weight. The purpose of this facility is to verify that the nodes along a route are actually online and initialize it by establishing a set of symmetric keys, one for each node on the route. These keys enable each node involved to check the header’s integrity, onion-decrypt the data payload, and retrieve the routing information. As processing Sphinx packets require expensive asymmetric cryptographic operations, it’s not used after the route is initialized.

This facility uses Sphinx[59] in the same way as described in HORNET[34] "4.3 Setup Phase" to which we refer you. Otherwise, like in TARANET[40], the expiration time EXP is not part of a Forwarding Segment FS, but include it alongside the FS in the packet (see TARANET Section 5.3.2). This allows the sender to set a different expiration time for each packet and limit the time window in which the packet is valid, which is necessary for replay protection.

To open a route, it requires the n-tuple representing the sequence of nodes from the sender node to the receiver one provided by the topology discovery facility.

Compared to TARANET, R-Stratum adds fields to advertise the intermediate node about the required service level agreement and to get in return one or more service levels offered.



which indicates the intended expiration time of the session

A relationship anonymous connection between a sender and a receiver require the source to establish a state between itself and every node on the path.

7.6 The flow setup process

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7.7 The data transmission process

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985 7.8 The validating facility

A node can decide to assume the role of validator of transactions (TX) on the network. In this case, it listens to two specific types of control plane flowlet that carry two types of information. The first contains the TX announcements that are collected in an in-memory pool (mempool). The second includes the certified bandwidths increase announcements to distribute the reward.

To describe the operation of this facility we differentiate the nodes into two types, the **VNodes**, in which the VFacility runs, that manage the function of identifying the topology within a certain distance from themselves, and the **nVNodes** that don't run it and delegate this function to another node.

995 This property derives from the fact that if two valid MW-Grin transactions are merged, then the result is another valid transaction.

In Bitcoin, a transaction is a collection of inputs and outputs and their respective signature. MW-Grin deviates from this model; a transaction consists of the following elements:

1. A set of **inputs**, that reference and spend a set of previous outputs.
- 1000 2. A set of new **outputs** that include:
 - (a) A value and a blinding factor (which is just a new private key) multiplied on a curve and summed to be validable by a range proof. (see Maxwell confidential transaction[60])
 - (b) A range proof (bulletproof[61]) that shows that the value is non-negative.
- 1005 3. A **transaction kernel** (<100 byte) that includes:
 - (a) An explicit **transaction fee**, in clear.
 - (b) A signature, computed by taking the **excess** blinding value (the sum of all outputs plus the fee, minus the inputs) and using it as a private key.

A key role is that of excess:

1010 *"rather than having each input sign the transaction, a sort of multi-signature (multisig) across all inputs, and all outputs is created. The sum of all the keys on all the outputs minus the sum of all the input keys is an elliptic curve point which can be treated as a public key. That public key is the multisig public key for all the transacting parties, the owner of every input and the owner of every output. And*
1015 *so a signature on this represents that the transaction was authorized by everybody involved. [...] The excess value is a multisig but it's also a proof that the transaction adds up to zero at the same time."*

So you can merge any two transactions, as long as you conserve both their kernel.

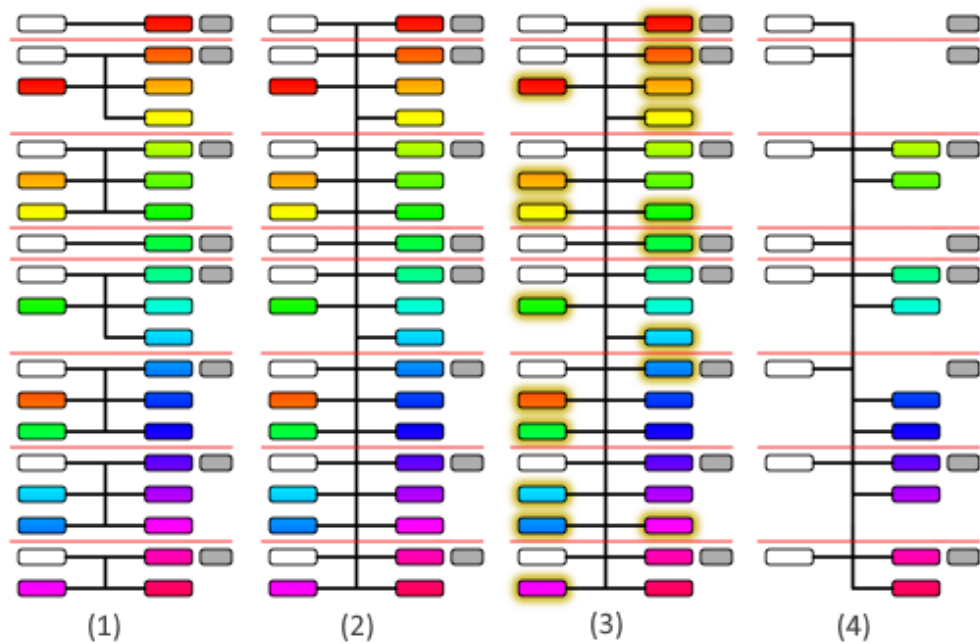


Figure 6. Pruning MimbleWimble, from Polstra A. slides[62] at Scaling Bitcoin 2016 conference in Milan: (1) The MW blockchain, (2) blocks are merged, (3) matching inputs and outputs selected and finally (4) removed. White box are coinbase transactions.

8 ONERARII STRATUM

1020 The goal of O-Stratum is to provide a transport protocol which cannot be filtered with Deep
Packet Inspection (DPI). To accomplish this intent, it must not be vulnerable to fingerprinting
using static string matching, packet length comparison, or timing profiling. Other attacks such
as coercion of operators are outside of the scope of the stack.

8.1 O-Stratum protocol data unit: Segment

1025 **Segment** is the name used uniquely for the PDU of O-Stratum inside the Nova stack.

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A ANNEXES

A.1 Candidate Cryptosystems

In this document, it's not necessary to make any final choice regarding cryptographic systems, but some points may already be highlighted.

Routers often embed part of the network protocols, adopting hardware solutions to accelerate the processing of the packets. Cryptography algorithms are exemplary candidates for this optimization. Therefore the design of a TCP-IP replacement poses the technological challenge compared to implementations that are thought to be only software regarding the hardware higher inertia to change.

Although one cryptographic system may reasonably be preferred today rather than another, no one can exclude that a vulnerability can be identified in the future. In particular, the approach of quantum computing represents a real threat that does not allow choices made now and forever.

For these reasons, the implementation of Nova will probably have to offer from the beginning at least two non-quantum resistant sets, each one containing for the same function a different cryptographic algorithm. The initial handshake of the related stratum communication let's two nodes agree which set to use.

Function	Set 1	Set 2
Key Exchange Scheme	DH	DH
Asymmetric Cryptography Algorithm	X25519	Secp256k1
Authenticated Encryption with Associated Data	AES_256_GCM[63]	CHACHA20_POLY1305[64]
Hash function	SHA3-256	BLAKE2s

Table 7. Nova suggested initial cryptographic sets

As the purpose of this paper is to demonstrate the consistency and feasibility of Nova stack, in the following of this paper we assume the use of "Set 1" for examples.

A.2 Why a new monetary unit

In Nova transactions are stored following the UTXO model also used by Bitcoin.

However Bitcoin cannot be used inside Nova for many reasons:

- an adversary by correlating the costs of each hop of routes that are in the clear in its blockchain can know the fact that in a given moment of time there was a communication between two parties,
- all transaction history is necessary to determine the correct current state of the chain (balance of the wallets). Given the number of nodes of the Internet and the number of packets sent, a traditional blockchain is not the solution,
- using the Proof of Work means giving up the possibility of rewarding with seigniorage ISPs that add bandwidth to the network. While it's true that PoW is a proven security model, it can become environmentally and therefore politically unsustainable[65] due to energy consumption. This is a problem regardless of the correctness of the academic analysis on the subject, the only possibility that it is on the table of political discussion, frame the topic in a debate in which it would be better not to be. In fact who has a position returns in the old business model, instead of competing in innovation, can operate on the political level through lobbying, riding an already strong political narrative.

- it has lousy scalability if used without resorting to solutions such as Lightning Network[66] (LN) which is an overlay network that relies on Bitcoin to:
 1. improves scalability because it consolidates transactions,
 2. make possible to do micropayments by reducing fees and overcoming dust limit, and
 3. permit instantaneous transactions enabling many use cases.

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Since the rationale behind this whitepaper is to build a stack from scratch to remedy the limitations of the decades-old TCP-IP with a network effect of billions of nodes, it makes no sense to approach scalability problems of a relatively recent Bitcoin that has only a few thousand of nodes adopting Bitcoin with a fixing layer like LN in Nova.

Nova is a network of nodes that route traffic, not funds. Unlike LN intermediate nodes, it don't have to provide liquidity to fund channels eliminating at the source the possibility that a router can be considered a money transmitter by regulators.

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We choose an alternative approach that solves these problems at the origin, adopting mimblewimble[53][54] a massively prunable blockchain technology. From the original idea of MW, two cryptocurrency are being developed, Grin[67] and Beam[68].

MW is particularly suitable because it use homomorphic encryption to hide transaction amount and most of the data of transactions whose output has already been spent can be purged out from the blockchain.

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Must include all authors, identified by initials, for example: A.A. conceived the experiment(s), A.A. and B.A. conducted the experiment(s), C.A. and D.A. analysed the results. All authors reviewed the manuscript.

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